Approximate feasibility can miss the intended constraint

One pair, one criterion; M = 1. Exact forward error = t + 1/10; approximate forward error = t.

t = 0; q = 1/20; ε = 1/10

q − ε = -1/20; q + ε = 3/20

Report	N	$\widehat{E}$	E	Inner	Exact	Nominal	Outer
Hold	0	0	0	**Yes**	**Yes**	**Yes**	**Yes**
A>B	1	0	1/10	No	No	**Yes**	**Yes**
B>A	1	1	9/10	No	No	No	No

t = 1/5; q = 1/4; ε = 1/10

q − ε = 3/20; q + ε = 7/20

Report	N	$\widehat{E}$	E	Inner	Exact	Nominal	Outer
Hold	0	0	0	**Yes**	**Yes**	**Yes**	**Yes**
A>B	1	1/5	3/10	No	No	**Yes**	**Yes**
B>A	1	4/5	7/10	No	No	No	No

Inner: $\widehat{E} \le (q-\varepsilon)N$ Exact: $E \le qN$

Nominal: $\widehat{E} \le qN$ Outer: $\widehat{E} \le (q+\varepsilon)N$

At t = 0, A>B fails the inner test 0 ≤ −1/20. Clipping the threshold to 0 wrongly admits it.
Its exact error 1/10 > q = 1/20 still violates the intended protection.

Hold (N = 0): all count rows read 0 ≤ 0; the displayed error fraction is undefined.
Exact refers to the intended supplied law; membership columns are separate tests.